

Received-Signal-Strength-Based Localization in Wireless Sensor Networks

This paper presents an overview of recent developments in received-signal-strength-based localization for wireless sensor networks, with an emphasis on minimizing the communication overheads.

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ABSTRACT | In this paper, an overview of recent developments in received-signal-strength (RSS)-based localization in wireless sensor networks is presented. Several important practical issues and their solutions are discussed. To **save communication bandwidth and sensor energy**, a maximum-likelihood estimator based on quantized data is presented along with its corresponding Cramér–Rao lower bound (CRLB) and optimal quantizer design schemes. **For further system resource savings, an iterative sensor selection approach is presented to activate only the most informative sensors**, by maximizing the mutual information or minimizing the posterior CRLB at each iteration. For a resource constrained WSN with imperfect wireless channels, channel-aware target localization is described, where the channel model is incorporated into the localization scheme itself, thereby improving performance without increasing communication overhead. Another practical issue involving the presence of malicious sensors called Byzantines is discussed and mitigation schemes are provided. A recent coding-theory-based approach which is both computationally inexpensive and robust to such malicious attacks is also discussed.

KEYWORDS | Byzantine attacks; channel-aware estimation; Cramér–Rao lower bound (CRLB); localization; quantization; received signal strength (RSS); security; sensor selection; wireless sensor networks

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I. INTRODUCTION

Localization is an important research problem in many systems such as radar, sonar, the global positioning system (GPS), wireless networks, cognitive radio networks, multimedia systems, and wireless sensor networks (WSNs). Source localization using a network of sensors or wireless nodes has far-reaching applications, e.g., wireless location services; search, rescue, and navigation in harsh environments; autonomous vehicles; Internet of Things; and monitoring and surveillance for security and defense. As a result, this research topic has received continuous interest and has been investigated widely [1]–[6].

For wideband wireless networks, localization performance limits in terms of Cramér–Rao lower bounds (CRLBs) were derived in [7] and [8] based on received signal waveforms. In [9]–[11], wideband ranging was investigated for harsh wireless environments with multipath fading and obstructed propagation. The problem of multinode or multisource localization in networks was investigated in [12]–[14]. Approaches for mobile target localization and tracking in a network setting were proposed based on particle filtering [15], [16], belief condensation filtering [17], [18], belief propagation [19], and message passing [20]. Closely related to network localization problems, network navigation approaches were proposed in [21]–[23] for mobile agents to estimate their own locations by sharing and fusing information in cooperative networks.

To achieve the best estimation performance under resource constraints, the sensor management problem has been studied extensively for localization and tracking [24]–[40]. Power allocation approaches were developed to achieve better performance for network localization [33] and network navigation [35]. A computational geometry

framework was proposed to provide the optimal node prioritization strategy for network localization [39]. Control of mobile agents in distributed networks has been investigated to achieve the maximum amount of information measured by mutual information [30], [32] or posterior entropy [36]. Optimal node placement for localization was studied in [37] and [38].

The focus of this paper is on localization in WSNs. Due to the tremendous advances in the theory and practice of networking, wireless communications, computing, micro-fabrication, and microprocessors over the last few decades along with significant cost reduction, practical deployment of WSNs for collaborative signal processing has become a reality. A WSN consists of a large number of sensors that are typically battery driven and hence operate on a tight energy budget. These sensors are typically low-cost, and have limited sensing and communication abilities.

WSNs are currently being used for a variety of monitoring and surveillance applications. These require the WSNs to perform inference tasks such as detection, localization, and tracking of objects and events as they unfold. In this paper, we mainly address location estimation of any source that emits energy which follows the power attenuation model (such as an acoustic source or an electromagnetic source). It is assumed that a point source emits signals omnidirectionally and the signals get attenuated inversely proportional to the distance from the source situated at the ground surface [41]–[47].

Several localization methods have been proposed by researchers in the past [45], [48]–[57]. In [45], [48], [49], and [53]–[55], direction-of-arrival (DOA)-based estimation has been considered for narrowband signal sources. On the other hand, time-difference-of-arrival (TDOA)-based estimation methods are more suitable for broadband signal sources [50]–[52], [56], [57]. A WSN typically consists of inexpensive sensors which are not necessarily fully synchronized. Hence, the TDOA-based method is not practical for such low-cost networks. However, the presence of densely deployed sensors in the region of interest (ROI) makes it possible to have accurate energy-based target localization. While received signal strength indicator (RSSI) measurements are commonly used for target detection, one can use them for localization without any additional sensor functionalities (DOA, TDOA, etc.). In this paper, we present an overview of RSSI-based methods for localization. We also discuss localization methods in the presence of practical impairments such as channel corruptions and Byzantine attacks.

Researchers have investigated energy-based localization in [41], [47], and [58]–[60]. Least square methods were proposed in [41] that use energy ratios between sensors to localize a single acoustic source. Using the same signal attenuation model, a maximum-likelihood (ML)-based acoustic source localization method was presented in [47]. In [59], the localization performance was analyzed for two common multilateration algorithms based on range estimates obtained from TOA or RSS measurements. A Bayesian approach based on particle filtering to fuse TOA and RSS measurements was developed in [60] for localization in harsh wireless environments.

Energy-based source localization is a nonlinear non-convex problem, and the ML estimator is computationally expensive when searching for the global maximum. Recently, there has been an increasing interest in relaxing the nonconvex problem, and localize the source through semidefinite programming (SDP) [61]–[63].

In addition to the signal-decay-model-based approaches, the so-called RSS fingerprinting techniques have been proposed to localize a wireless emitter [64], [65]. In RSS fingerprinting, during an offline survey phase, wireless RSS data for various possible target locations are obtained to build a radio map database. Using unique fingerprints for different target locations, the emitter location can be estimated during the online phase. In [66], for indoor localization, the phase information of the channel state information (CSI) was collected for location fingerprinting instead of the RSS measurements, and a deep learning approach was applied during the offline training stage.

In [41], [47], and [58]–[65], the source location was estimated using analog measurements from the sensors. Transmission of analog data is expensive and consumes a large amount of system resources, such as energy and bandwidth. One potential solution to reduce this communication overhead is for local sensors to send only binary or multibit quantized data to the processing node (fusion center). Target localization methods have been developed at the fusion center (FC) based on quantized sensor data [67]–[71]. To further conserve system resources, such as energy and bandwidth, an iterative sensor selection approach for localization was developed in [72] to activate only the most informative sensors, which achieves a better tradeoff between energy efficiency and estimation performance.

In practice, the communication channels between the sensors and the FC are nonideal and subject to noise and other channel impairments such as fading. In [44]–[47] and [73], the signals received by the FC are modeled as analog measurements corrupted by additive white Gaussian noise (AWGN). This assumption does not accurately represent data transmission in practical digital communication systems. For realistic multipath fading channels in digital communication systems, the wireless channel imperfections are mitigated by the use of error-correction codes. While it is a plausible solution, it adds additional communication overhead causing increased consumption of energy and bandwidth. In [74], the authors developed an efficient channel-aware localization algorithm that incorporates the communication channel models.

Recently, security of information networks has become a major concern. In this context, secure localization is crucial as potential malicious entities may attempt to disrupt the network, diminish its capability, and potentially misuse the information to carry out subversive activities. Only in the recent past, researchers investigated the problem of security threats [75]–[78] on sensor networks as they carry out inference tasks. In this paper, we focus on one particular class of security attacks, known as the Byzantine data attack [79], [80] (also referred to as the data falsification attack). A Byzantine attack involves malicious sensors within the

network which send false information to the FC to disrupt the global inference process. In [81], the effect of Byzantines on the overall estimation performance is characterized in terms of the optimal attack strategies. Game-theoretic formulations were explored to analyze two-player interactions. To address the problem from the system's perspective, two types of Byzantine mitigation schemes have been considered in the literature: Byzantine identification schemes and Byzantine tolerant schemes. In [81], using learning-based techniques, Byzantine identification schemes have been designed that learn the identities of Byzantines in the network and use this information to improve system performance. When such schemes are not possible, Byzantine tolerant schemes using error-correcting codes have been developed [82] that tolerate the effect of Byzantines and maintain good performance in the network. This helps mitigate the impact of Byzantines and channel impairments, while also resulting in a computationally inexpensive localization scheme.

The rest of the paper is organized as follows. In Section II, the unified system model considered throughout the paper and some background on estimation theory are presented. In Section III, the ML location estimator based on quantized data is presented and its performance limit is characterized by the CRLB. Optimal quantizer design at local sensors is discussed in Section IV. An energy-efficient iterative target localization scheme is presented in Section V. To address practical concerns, in Sections VI and VII, the scenarios where imperfect channels and malicious sensors exist in the WSN are considered, respectively. In Section VIII, a recent coding-based localization scheme in WSNs is presented, which is energy efficient, and robust to imperfections due to channel errors and malicious sensors. The paper concludes with some discussion in Section IX.

II. SYSTEM MODEL AND BACKGROUND

In this section, we provide the system model for source localization using RSSI measurements and a brief introduction to fundamental estimation theory.

A. Signal Model

In this paper, the problem of estimating target location based on RSSI measurements is considered, in which it is assumed that the signal intensity attenuates as the distance from the target increases in a sensor field with N sensors. Therefore, the signal intensity contains the range information of the target.

Here, an isotropic signal attenuation model is adopted

$$a_i^2 = \frac{G_i P'_0}{\left(\frac{d_i}{d_0}\right)^n} \quad (1)$$

where a_i is the signal amplitude at the i th sensor, G_i is the gain of the i th sensor, P'_0 is the emitted power of the target

measured at a reference distance d_0 , and d_i is the Euclidean distance between the target and the i th sensor

$$d_i = \sqrt{(x_i - x_t)^2 + (y_i - y_t)^2} \quad (2)$$

in which (x_i, y_i) and (x_t, y_t) are the coordinates of the i th sensor and the target, respectively, and n is the power decay exponent. For simplicity, in this paper, the power decay exponent n is assumed to be known and identical for all the sensors. An unknown but identical decay exponent can be jointly estimated with the source location as proposed in [83]. Unknown and different decay exponents for different sensors can be dynamically estimated from RSS measurements as shown in [84]. Additionally, RSS fingerprinting techniques can be applied to address the issue of nonisotropic signal attenuation.

For simplicity, it is assumed that $G_i = G$ for $i = 1, \dots, N$, $P_0 = G P'_0$, and $d_0 = 1$ m. As a result, (1) becomes

$$a_i^2 = \frac{P_0}{d_i^n}. \quad (3)$$

It is assumed that all the sensors are in the far field, and the target is at least d_0 m away from any sensor all the time.

The intensity decay model defined here is quite general. It encapsulates several cases including spherical acoustic waves radiated by a simple source propagating through the air, when the intensity of the waves decays at a rate inversely proportional to the square of the distance [85]. Equation (3) is also a widely used model for isotropically radiated electromagnetic waves propagating in free space [86].

It is further assumed that the average signal amplitude at the i th sensor (obtained by averaging over multiple samples) is corrupted by additive Gaussian noise

$$s_i = a_i + \omega_i \quad (4)$$

where $\omega_i \sim \mathcal{N}(0, \sigma^2)$ for $i = 1, \dots, N$. At each sensor, the signal amplitude s_i is quantized into a discrete value m_i and transmitted to the processing node (FC).

B. Background on Estimation Theory

Since localization is a nonlinear estimation problem, in this section, we provide a brief introduction to estimation theory, to make the paper more self-contained.

Assuming that a sensor makes a nonlinear measurement (z) of the system parameter θ , in the presence of random noise, the estimator aims to find an estimate of θ based on the measurement z , which can be denoted as $\hat{\theta}(z)$.

For a fixed but unknown parameter, a non-Bayesian estimator can be used to give the estimate of the parameter. A well-known non-Bayesian estimator is the ML estimator, which maximizes the likelihood function: $\arg \max_{\theta} p(z|\theta)$. For any unbiased non-Bayesian estimator, its covariance matrix is bounded by the CRLB [87]

$$E\{[\hat{\theta}(z) - \theta][\hat{\theta}(z) - \theta]^T\} \geq J^{-1} \quad (5)$$

in which the inequality sign is defined in the positive-semidefinite sense. $\mathbf{J}$ is the Fisher information matrix (FIM), whose inverse provides the CRLB matrix, and

$$\mathbf{J} = E[-\Delta_{\theta}^{\theta} \log p(\mathbf{z}|\theta)] \quad (6)$$

where $\Delta_{\theta}^{\theta} = \nabla_{\theta} \nabla_{\theta}^T$ is the second derivative (Hessian) operator, and ∇ is the gradient operator.

For a random parameter with prior $p(\theta)$, two widely used Bayesian estimators are the maximum *a posteriori* (MAP) estimator, which maximizes the posterior probability density function (pdf) $\arg \max_{\theta} p(\theta|\mathbf{z})$, and the minimum mean squared error (MMSE) estimator, which is also the conditional mean $E\{\theta|\mathbf{z}\}$. For Bayesian estimation problems, the posterior CRLB (PCRLB) [87], [88], or the Bayesian CRLB, provides a lower bound on the estimation mean squared error (MSE)

$$E\{[\hat{\theta}(\mathbf{z}) - \theta][\hat{\theta}(\mathbf{z}) - \theta]^T\} \geq \mathbf{F}^{-1} \quad (7)$$

where $\mathbf{F}$ is the Bayesian FIM

$$\mathbf{F} = E[-\Delta_{\theta}^{\theta} \log p(\theta, \mathbf{z})]. \quad (8)$$

Note the difference between (6) and (8). In the former, the pdf used is the likelihood function $p(\mathbf{z}|\theta)$, while in the latter the joint pdf $p(\mathbf{z}, \theta)$ is employed. Similarly, the expectations in these two equations are with respect to $p(\mathbf{z}|\theta)$ and $p(\mathbf{z}, \theta)$, respectively.

III. ML LOCALIZATION AND ITS PERFORMANCE LIMIT

In this section, we present the ML location estimator in WSNs based on quantized data, and compare it with localization approaches using analog data, in terms of both the MSE and the CRLB.

A. Location Estimation Using Analog Data

In this section, it is assumed that the unknown parameter to be estimated is $\theta = [P_0, x_t, y_t]^T$. First, consider the case in which sensors transmit analog data to the FC, where they are used to estimate the target location. The analog sensor data are denoted as $\mathbf{s} = [s_1, \dots, s_N]^T$. Due to the additive Gaussian noise assumption, up to an additive constant, the log-likelihood function can be written as

$$\log p(\mathbf{s}|\theta) = -\frac{1}{2} \sum_{i=1}^N [s_i - a_i(\theta)]^2. \quad (9)$$

The ML estimate is determined by finding the θ that maximizes the log-likelihood

$$\arg \max_{\theta} \log p(\mathbf{s}|\theta) \Leftrightarrow \arg \min_{\theta} \sum_{i=1}^N [s_i - a_i(\theta)]^2 \quad (10)$$

The CRLB, which provides a fundamental performance limit for any unbiased non-Bayesian estimator, is derived for this problem and is provided in the following theorem [68].

Theorem III.1: For the localization problem based on analog data, the FIM $\mathbf{J}$, the inverse of the CRLB matrix, is

$$\mathbf{J} = \frac{1}{\sigma^2} \sum_{i=1}^N \mathbf{C}_i$$

where

$$\mathbf{C}_i = \begin{bmatrix} \frac{1}{4 P_0 d_i^n} & \frac{n(x_i - x_t)}{4 d_i^{n+2}} & \frac{n(y_i - y_t)}{4 d_i^{n+2}} \\ \frac{n(x_i - x_t)}{4 d_i^{n+2}} & \frac{P_0 n^2 (x_i - x_t)^2}{4 d_i^{n+4}} & \frac{P_0 n^2 (x_i - x_t)(y_i - y_t)}{4 d_i^{n+4}} \\ \frac{n(y_i - y_t)}{4 d_i^{n+2}} & \frac{P_0 n^2 (x_i - x_t)(y_i - y_t)}{4 d_i^{n+4}} & \frac{P_0 n^2 (y_i - y_t)^2}{4 d_i^{n+4}} \end{bmatrix}.$$

B. Location Estimation Using Quantized Data

In WSNs, due to limited communication bandwidth and limited sensor battery power, it is desirable for the sensors to transmit quantized data instead of analog data to the FC. In [68], the location estimation problem using quantized data was investigated. Assume that each sensor sends quantized M -bit data to the FC, which are collectively denoted as $\mathbf{m} = [m_1, \dots, m_N]^T$, where m_i can take any discrete value from 0 to $2^M - 1$. To simplify the notation, L is defined as $L = 2^M$. The quantization process for the i th sensor is such that

$$m_i = \begin{cases} 0 & -\infty < s_i < \eta_{i1} \\ 1 & \eta_{i1} \leq s_i < \eta_{i2} \\ \vdots & \vdots \\ L-1 & \eta_{i(L-1)} \leq s_i < \infty \end{cases} \quad (11)$$

where $\eta_i = [\eta_{i1}, \dots, \eta_{i(L-1)}]^T$ is the set of quantization thresholds used at the i th sensor.

Due to the Gaussian noise assumption, the probability that m_i takes a specific value l is

$$p_{i,l}(\theta) = Q\left(\frac{\eta_{il} - a_i(\theta)}{\sigma}\right) - Q\left(\frac{\eta_{i(l+1)} - a_i(\theta)}{\sigma}\right) \quad (12)$$

where $Q(\cdot)$ is the complementary cumulative distribution function of the standard Gaussian distribution.

The FC estimates the unknown parameter θ from the data $\mathbf{m}$, using the log-likelihood function at the FC which is given by

$$\log p(\mathbf{m}|\theta) = \sum_{i=1}^N \log p_{i,m_i}(\theta). \quad (13)$$

ML estimation is now the following optimization problem: $\max_{\theta} \log p(\mathbf{m}|\theta)$. The CRLB for this estimation problem has been derived and stated in the following theorem.

Theorem III.2: Denoting the FIM for a location estimator based on quantized data as $\mathbf{J}$, its elements are as follows [68]:

$$\begin{aligned} J_{11} &= \sum_i \kappa_i d_i^{-2n} / a_i^2 \\ J_{12} &= J_{21} = n \sum_i \kappa_i d_i^{-(n+2)} (x_i - x_t) \\ J_{13} &= J_{31} = n \sum_i \kappa_i d_i^{-(n+2)} (y_i - y_t) \end{aligned}$$

$$\begin{aligned} J_{22} &= n^2 \sum_i \kappa_i a_i^2 d_i^{-4} (x_i - x_t)^2 \\ J_{23} &= J_{32} = n^2 \sum_i \kappa_i a_i^2 d_i^{-4} (x_i - x_t)(y_i - y_t) \\ J_{33} &= n^2 \sum_i \kappa_i a_i^2 d_i^{-4} (y_i - y_t)^2 \end{aligned} \quad (14)$$

where

$$\kappa_i = \frac{1}{8\pi\sigma^2} \sum_{l=0}^{L-1} \frac{\lambda_{il}^2}{p_{il}(\eta_i, \theta)}$$

and

$$\lambda_{il} = e^{-\frac{(\eta_{il} - a_i)^2}{2\sigma^2}} - e^{-\frac{[\eta_{i(l+1)} - a_i]^2}{2\sigma^2}}.$$

We now present simulation results for the ML estimator discussed in [68]. The ML estimator is compared with several heuristic estimation methods, which are based on weighted averages (WA) of all the sensors' locations. The first two WA methods (WA-I and WA-II) have i th sensor's weight proportional to the l th power of the data (m_i^l , for $l = 1, 2$, respectively). The third WA method (WA-III) considers only a subset of sensors, in which only sensors whose data have the highest value of m_i get equal nonzero weights. Note that a drawback of these methods is that they cannot estimate P_0 . In Fig. 1, the performance of different estimators is plotted as a function of $K = \sqrt{N}$. From this figure, it can be seen that the ML method has superior performance than the heuristic methods. Also, as the number of sensors increases, the optimal ML estimator converges to the CRLB limit quickly.

In Fig. 2, the ML estimator using quantized data is compared with that using analog data. As we can see, the root mean square (RMS) estimation errors obtained from Monte Carlo trials for the ML estimator using analog data are indistinguishable from their CRLBs. ML estimator's performance

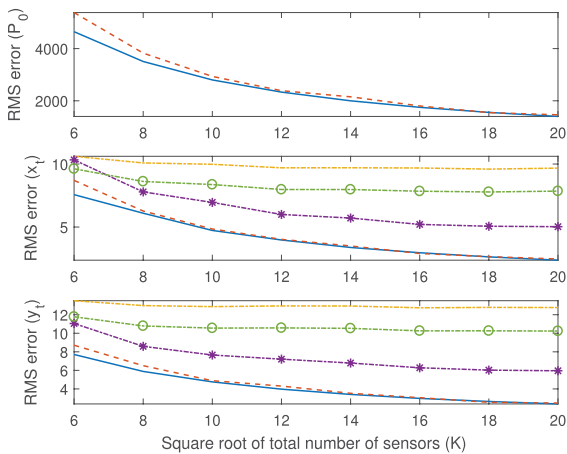


Fig. 1. Root mean square (RMS) errors of different estimation methods ($n = 2$, $P_0 = 25,000$, $x_t = 15$ m, $y_t = 20$ m, $M = 2$, $\sigma = 1$). RMS estimation errors for x_t and y_t are in meters. Solid line: CRLB; dashed line: ML; dashed dotted line: WA-I; dashed dotted line + circle: WA-II; dashed dotted line + star: WA-III. Quantization threshold $\eta = [0.82, 1.70, 2.72]$ [68].

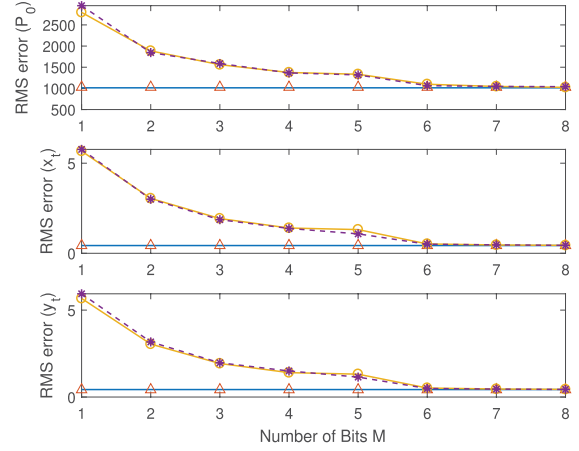


Fig. 2. RMS errors for ML estimators using quantized data and analog data ($x_t = y_t = 0$ m, $N = 196$). Solid line: CRLB for analog data; triangle: ML estimator using analog data; solid line + circle: CRLB for quantized data; dashed line + star: ML estimator using quantized data. Quantization thresholds are calculated using the Fisher-information-based heuristic quantization (FIHQ) method, which is described in Section IV [68].

is close to the limit (CRLB) even with quantized data, and this performance converges to the performance using analog data as the number of bits increases. This is expected as the precision loss due to quantization reduces as the number of bits increases, which is almost negligible when 6 bits are used.

IV. OPTIMAL QUANTIZATION THRESHOLDS

In the previous section, we presented the performance limit of location estimation using quantized data. A key aspect to notice is that the performance (FIM and CRLB) are functions of the thresholds used for quantization (ref. Theorem III.2). Hence, the optimal thresholds for quantization can be designed by optimizing CRLB or FIM (or their functions such as trace or determinant). For a large sensor network, this is a large-scale optimization problem due to the total number of parameters to be optimized. For example, when 64 sensors in the network perform 10-bit quantization, there are 65,472 variables over which the optimization problem needs to be solved. For theoretical analysis and in practical implementation, let us assume that all the sensors use identical thresholds η . Then, the number of parameters of the optimization problem reduces to just $L - 1$ variables.

As Theorem III.2 states, the optimization cost function depends on the unknowns, signal power P_0 , and target location (x_t, y_t) , which are to be estimated. Besides these unknowns, it also depends on the sensor locations, which are uncertain as the sensors are randomly deployed in many WSN applications. Therefore, one cannot set the optimal thresholds before sensor deployment due to the uncertainty in sensor locations. Two approaches have been considered

in [68] namely, entropy-based heuristic quantization (EHQ) and Fisher-information-based heuristic quantization (FIHQ), to design quantization thresholds. They do not require the above mentioned prior knowledge about target parameter and sensor locations, and are simple to calculate and robust in various scenarios.

To implement EHQ or FIHQ, let us consider the distribution of a_i . From (2) and (3), signal amplitude is a function of P_0 , (x_i, y_i) , and (x_t, y_t) . Assuming these parameters follow a certain distribution, we can derive the distribution of a_i . Let P_0 follow a uniform distribution in $[0, P_m]$ and x_i, x_t, y_i, y_t are independent identically distributed (i.i.d.) random variables following a uniform distribution in the interval $[-b/2, b/2]$, where b is the length of the ROI. Also, assume that all the sensors are at a distance greater than d_0 from the target, and the power decay exponent is $n = 2$. Given these assumptions, the pdf of a_i measured at any random sensor location is given by [68]:

Theorem IV.1: The pdf of the signal amplitude measured at a random location, denoted as z , is

$$p(z) = \frac{2}{\alpha P_m} \times \begin{cases} \left(\frac{b^2}{3} - \beta\right)z & 0 < z \leq \frac{\sqrt{P_m}}{\sqrt{2}b} \\ \left[\frac{b^2}{15} - \beta + g\left(\frac{P_m}{z^2}\right)\right]z & \frac{\sqrt{P_m}}{\sqrt{2}b} < z \leq \frac{\sqrt{P_m}}{b} \\ \frac{P_m^3}{3b^4z^5} - \frac{8P_m^2}{5b^3z^4} + \frac{\pi P_m^2}{2b^2z^3} - \beta z & \frac{\sqrt{P_m}}{b} < z \leq \frac{\sqrt{P_m}}{d_0} \\ 0 & \text{o.w.} \end{cases} \quad (15)$$

where

$$\alpha = 1 + \frac{8d_0^3}{3b^3} - \pi \frac{d_0^2}{b^2} - \frac{d_0^4}{2b^4} \quad (16)$$

$$\beta = \frac{\pi}{2b^2}d_0^4 + \frac{d_0^6}{3b^4} - \frac{8d_0^5}{5b^3} \quad (17)$$

and the function $g(\cdot)$ is defined as

$$g(t) = \frac{t^2}{b^2} \arcsin\left(\frac{2b^2}{t} - 1\right) - \frac{t^2}{b^2} - \frac{t^3}{3b^4} + \frac{2}{15b^3} \sqrt{t - b^2} (12t^2 + b^2t + 2b^4). \quad (18)$$

Based on this distribution of the signal amplitude $z = a_i$, the probability that the quantization process has an output symbol l ($l = 0, \dots, L-1$) is [cf. (12)]

$$P_l = \int_0^{\frac{\sqrt{P_m}}{d_0}} \left[Q\left(\frac{\eta_l - z}{\sigma}\right) - Q\left(\frac{\eta_{l+1} - z}{\sigma}\right) \right] p(z) dz. \quad (19)$$

The EHQ method builds on the observation that the quantization threshold, to convey any relevant information, should not be too high or too low. This leads to a quantizer that maximizes the entropy of the symbols sent by the sensors. The maximum entropy can be achieved by choosing a

set of thresholds so that the probabilities of each symbol are the same, namely $P_l = 1/L$, ($l = 0, \dots, L-1$). For $L = 2$, the probability that all the sensors send the same symbols is significantly reduced.

Since all the information about the unknown θ is contained in sensors' signal amplitudes (a_i 's), if one can accurately recover all the signal amplitudes from the corresponding quantized data m_i 's ($i = 1, \dots, N$) received at the FC, an accurate estimate of θ can be obtained. It is not difficult to derive the Fisher information about a_i contained in m_i

$$F(\eta) = \int_0^{\frac{\sqrt{P_m}}{d_0}} 4\kappa(\eta, z)p(z)dz \quad (20)$$

where $\kappa(\eta, z)$ has been defined in Theorem III.2. The threshold η can then be designed such that in data $\mathbf{m}$, there is maximum amount of Fisher information about z : $\arg \max_{\eta} F(\eta)$.

The choice of thresholds is critical especially when binary quantization is used. In the following example, the "optimal" thresholds are designed assuming $\theta = [25,000 \ 0 \ 0]^T$. In reality, there is a mismatch in P_0 . In Table 1, the effect of using the "mismatched" optimal threshold on the CRLB performance of the estimator, and those of using heuristic thresholds for varying P_0 's are shown. As we can see, the optimal threshold gives a poor performance for "mismatched" P_0 values. For example, when $P_0 = 200$, the optimal threshold leads to an ill-conditioned estimation problem and gives rise to an estimation error of $\sqrt{V} = 1.25 \times 10^{12}m$. This is because the optimal thresholds, which are designed to match a greater P_0 , are too high in this case. The FC receives hardly any 1's. On the other hand, the proposed EHQ and FIHQ methods need minimal prior system information, but still have robust performance, even for extreme values of P_0 . Note also the case of $P_0 = 20,000$ where the mismatch is not significant, and the "optimal" threshold outperforms those of the EHQ and FIHQ.

V. ENERGY-EFFICIENT ITERATIVE TARGET LOCALIZATION

For the location estimation approach discussed using quantized data, simultaneous transmission of quantized data from all the sensors to the FC is needed which introduces some practical challenges. First, not all the sensors have useful information as the sensors that are far away from the target have little useful information to provide while still consuming energy for data transmission. Second, an independent channel is needed for each sensor's data transmission. Instead of using data from all the sensors, an energy-efficient iterative sensor selection approach for source localization was presented in [72].

Table 1 Square Root of Location Variance V

P_0	200	4000	20000	50000
Optimal threshold	1.25×10^{12}	1.47×10^5	2.30	105.26
EHQ	18.43	10.17	9.89	11.65
FIHQ	17.45	11.02	10.94	13.40

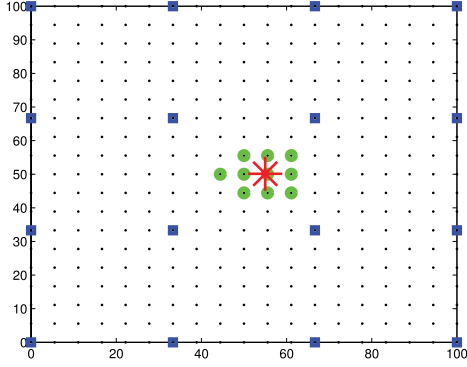


Fig. 3. WSN model (black points: sensor locations; blue squares: anchor sensors; green circles: activated sensors after ten iterations; red star: target, $A = 1$).

A. Iterative Sensor Activation for Target Localization

The energy-efficient iterative sensor selection approach is illustrated in Fig. 3. The algorithm starts with a small number (N_0) of anchor sensors sending their local M -bit data to the FC, which uses it to obtain a coarse location estimate. After getting an initial estimate, a few (A) non-anchor sensors that are deemed to be most informative in providing an estimate are activated. These sensors send their local M -bit quantized data to the FC which refines the location estimate accordingly. This continues in an iterative manner. Let $\mathbf{m}_{0:i} = [m_1, m_2, \dots, m_{N_0+iA}]^T$ denote the collected sensor data at the i th iteration where $i \in \{0, 1, \dots\}$ and A denotes the number of sensors activated at each iteration. In this section, $\theta = [x_t, y_t]^T$ is treated as a random parameter which has a certain prior pdf. The posterior pdf of the source location is approximated using an importance-sampling-based Monte Carlo method similar to that presented in [89]

$$p(\theta | \mathbf{m}_{0:i}) = \sum_{j=1}^{N_s} w^{j,i} \delta(\theta - \theta^{j,i}) \quad (21)$$

where N_s is the number of random samples (particles), $\theta^{j,i}$ denotes the j th particle at the i th iteration, and $w^{j,i}$ is the corresponding weight. The MMSE of the source location then has the form

$$\hat{\theta}_i = \sum_{j=1}^{N_s} w^{j,i} \theta^{j,i}. \quad (22)$$

Approaches for solving sensor management problems for target localization/tracking largely fall into two categories. The first is based on the CRLB [33], [35], [38], [39] or the PCRLB [27]–[29], [31] and the second is based on information-theoretic measures, such as entropy, mutual information, and Rényi divergence [24]–[26], [30], [32], [36], [90]–[93]. As shown later in the paper, one drawback of the information-theoretic measures is their high computational complexity, especially when the number (A) of sensors to be selected is large. In this section, two approaches are compared for sensor selection, based on mutual information and PCRLB, respectively.

Let $S^{i,v}$ be the set of A nonanchor sensors activated at the i th iteration according to the sensor selection strategy v . Then, there are $C_A^{N-N_0-(i-1)A}$ distinct A -element subsets of $N - N_0 - (i - 1)A$ remaining nonanchor sensors at the i -th iteration.¹ Let $\mathbf{m}^{i,v} = [m_1^{i,v}, \dots, m_A^{i,v}]^T$ denote the quantized measurements of $S^{i,v}$. At the i th iteration, S^{i,v^*} minimizes a certain cost function $\Psi_i(\cdot)$ as

$$v^* = \arg \min_v \Psi_i(v), \quad v \in \{1, 2, \dots, C_A^{N-N_0-(i-1)A}\}. \quad (23)$$

In this work, $\Psi_i(v)$ is selected as either the negative of the mutual information between the target location and the sensors to be selected, or the trace of the PCRLB matrix.

B. Evaluation of Mutual Information

Let $I(\theta, \mathbf{m}^{i,v}) = H(\mathbf{m}^{i,v}) - H(\theta | \mathbf{m}^{i,v})$ denote the mutual information between the source location and the measurements of the activated sensors. The entropy of the selected sensor data $H(\mathbf{m}^{i,v})$ can be derived as [72]

$$-\sum_{l_1=0}^{L-1} \dots \sum_{l_A=0}^{L-1} \left[\sum_{j=1}^{N_s} w^{j,i-1} \prod_{k=1}^A p(m_k^{i,v} = l_k | \theta^{j,i-1}) \right] \times \log \left[\sum_{j=1}^{N_s} w^{j,i-1} \prod_{k=1}^A p(m_k^{i,v} = l_k | \theta^{j,i-1}) \right] \quad (24)$$

in which $p(m_k^{i,v} = l_k | \theta^{j,i-1})$ has been provided in (12). The conditional entropy $H(\theta | \mathbf{m}^{i,v})$ is

$$-\sum_{k=1}^A \sum_{l=0}^{L-1} \sum_{j=1}^{N_s} w^{j,i-1} p(m_k^{i,v} = l | \theta^{j,i-1}) \log p(m_k^{i,v} = l | \theta^{j,i-1}). \quad (25)$$

C. Evaluation of Conditional PCRLB

Let $p(\mathbf{m}^{i,v}, \theta | \mathbf{m}_{0:i-1})$ denote the conditional joint pdf of the pair of $(\mathbf{m}^{i,v}, \theta)$ conditioned on the accumulated sensor data up to the $(i - 1)$ th iteration. Then, based on [88] and [94], the conditional PCRLB (C-PCRLB) on the MSE can be derived as follows:

$$E \{ [\hat{\theta}(\mathbf{m}_{0:i}) - \theta] [\hat{\theta}(\mathbf{m}_{0:i}) - \theta]^T | \mathbf{m}_{0:i-1} \} \geq \mathbf{F}^{-1} \quad (26)$$

where $\mathbf{m}_{0:i} = [\mathbf{m}_{0:i-1}^T (\mathbf{m}^{i,v})^T]^T$, and

$$\mathbf{F} = E \left[-\Delta_{\theta}^2 \log p(\mathbf{m}^{i,v}, \theta | \mathbf{m}_{0:i-1}) \right] \quad (27)$$

is the 2×2 conditional Bayesian FIM, whose inverse provides the conditional PCRLB matrix. Note that all the expectations in this section are with respect to $p(\mathbf{m}^{i,v}, \theta | \mathbf{m}_{0:i-1})$. Using the equality $p(\mathbf{m}^{i,v}, \theta | \mathbf{m}_{0:i-1}) = p(\mathbf{m}^{i,v} | \theta) p(\theta | \mathbf{m}_{0:i-1})$, an alternative expression for the Bayesian FIM can be written as

$$\mathbf{F} = E \left[-\Delta_{\theta}^2 \log p(\mathbf{m}^{i,v} | \theta) \right] + E \left[-\Delta_{\theta}^2 \log p(\theta | \mathbf{m}_{0:i-1}) \right] = \sum_{j=1}^{N_s} w^{j,i-1} \mathbf{J}(\theta^{j,i-1}) + \mathbf{F}_i. \quad (28)$$

¹ $C_k^n = n!/(k!(n-k)!)$ represents the binomial coefficient, which is the number of combinations of k items chosen from n items.

Table 2 Mean CPU Times of MI- and PCRLB-Based Sensor Selection Methods [72]

	PCRLB based Sensor Selection	MI based Sensor Selection
$A = 1$	0.26 s.	0.17 s.
$A = 2$	0.51 s.	9.49 s.
$A = 3$	0.75 s.	371.79 s.
$A = 4$	1.00 s.	15544.00 s.

In the first term in (28)

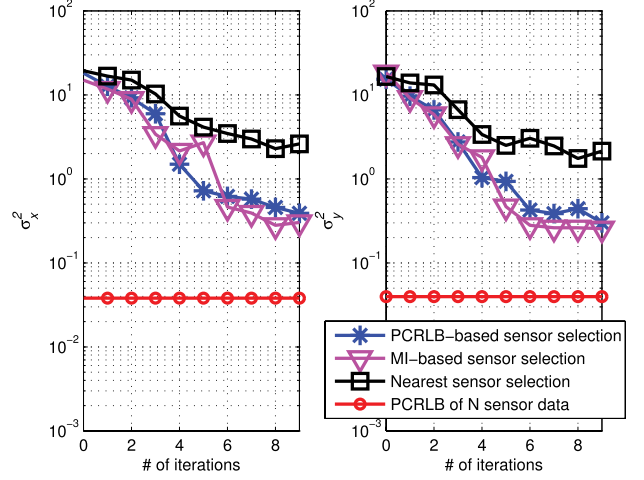
$$\begin{aligned} J(\theta^{j,i-1}) = & - \sum_{k=1}^A \sum_{l=0}^{L-1} \Delta_{\theta}^l \log p(m_k^{i,v} = l | \theta^{j,i-1}) \\ & \times p(m_k^{i,v} = l | \theta^{j,i-1}) \end{aligned} \quad (29)$$

is the standard FIM, which is provided by the lower right 2×2 submatrix of the FIM in Theorem III.2, and is further averaged over the prior pdf of the source location. The second term in (28) represents the *a priori* information, which can be approximated by a numerical approach proposed in [72].

Note that the mutual information requires $N_s \times L^A + A \times L \times N_s$ summations. In comparison, the Bayesian FIM function requires $A \times L \times N_s$ summations.

In Table 2, the average CPU times to evaluate mutual information (MI) and PCRLB are shown. From this table, it is clear that for $A \geq 2$, the MI-based methods have computational complexity that increases exponentially with A as L^A , while the C-PCRLB method has computation time that increases linearly with A as $L \times A$.

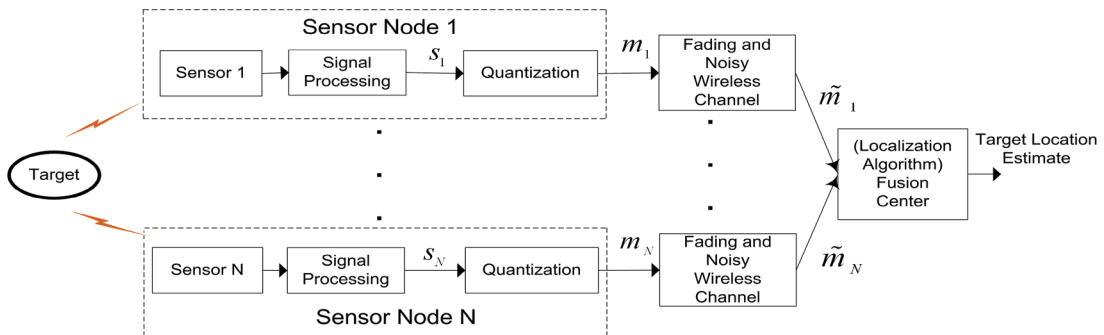
In Fig. 4, the estimation MSEs for MI- and C-PCRLB-based sensor selection methods are presented. The MSEs are also compared with the PCRLB found when all the $N = 361$ sensors send their M -bit quantized data to the FC. A third approach was considered, where A sensors nearest to the current location estimate are activated at every iteration. As can be seen in Fig. 4, the MI- and PCRLB-based sensor selection schemes have better MSEs in comparison to the nearest sensor selection scheme.


Fig. 4. Position MSEs for different sensor selection approaches at each iteration with $M = 3$ and $A = 1$ [72].

VI. CHANNEL-AWARE TARGET LOCALIZATION IN WSN

In the preceding section, we have considered the cases when the communication channels between the sensors and the FC are perfect. This may not be the case in practice, since line-of-sight between sensors and the FC may not exist. As mentioned before, error-correction coding could be used to mitigate wireless channel imperfections, but it results in additional communication overhead.

As an alternative, in [74], channel-aware localization algorithms were proposed that incorporate the imperfect nature of communication channels, as illustrated in Fig. 5. Sensors make their observations, which are then quantized locally before being transmitted to the FC via noisy channels. The FC, upon receiving these noisy sensor measurements, estimates the target location using a channel-aware localization algorithm. The issue of channel-aware inference was also considered in the context of detection problems (see [95] and the references therein).


Fig. 5. Generic channel-aware system model. s_i : raw sensor measurement; m_i : quantized sensor measurement; $\tilde{m}_i$: channel-corrupted sensor measurement.

In this section, different decoding strategies at the FC are analyzed by assuming a fixed channel model. There are two typical link layer designs that can be used at the FC depending on the decoding scheme: hard decoding and soft decoding. In hard decoding, the FC makes a hard decision for each noisy measurement and then performs the localization tasks, while in soft decoding, the FC directly uses the noisy measurements from sensors to estimate the location.

Let $\mathbf{y} = [\tilde{m}_1, \dots, \tilde{m}_N]^T$ be the noisy observation vector at the FC, where $\tilde{m}_i$'s represent the noisy versions of the quantized sensor local measurements. The FC estimates the source location based on $\mathbf{y}$.

Note that $\tilde{m}_i$ could be discrete or continuous, depending on the channel model and the decoding strategies. Without loss of generality and for simplicity, consider a binary quantization case with $L = 2$. Based on the fact that θ , m_i , and $\tilde{m}_i$ form a Markov chain, and the independent channel assumption, it can be shown that the log-likelihood function is [74]

$$\log p(\mathbf{y}|\theta) = \sum_{i=1}^N \log \left[\sum_{m_i} p(\tilde{m}_i|m_i) p(m_i|\theta) \right]. \quad (30)$$

A. Hard-Decoding Binary Channel

For hard-decoding strategies, we characterize the channel using a general binary channel (BC) model with bit error probabilities q_0 and q_1 , as shown in Fig. 6. For simplicity, assume that all the channels between the sensors and the FC are statistically independent and identical.

Using the channel model shown in Fig. 6, the probability distribution of the received sensor observation $\tilde{m}_i$, given the target location, is

$$p(\tilde{m}_i = m_i|\theta) = \sum_{m_i=0}^1 p(\tilde{m}_i = m_i|m_i) p(m_i|\theta) \quad (31)$$

Note that the first term in the summation in (31) represents the channel statistics. For example, given a target

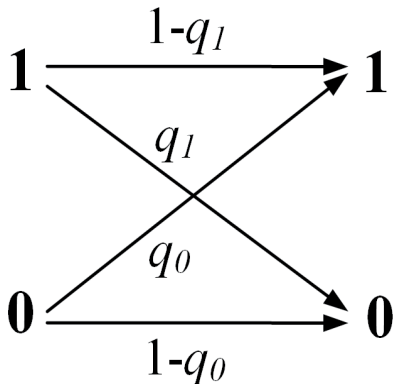


Fig. 6. A general binary channel model.

location θ , the probability of a received observation from sensor i being equal to 1 is $p(\tilde{m}_i = 1|\theta) = p(m_i = 1|\theta)(1 - q_1) + p(m_i = 0|\theta)q_0$.

Note that the FC only needs to know the received signal decay model, the sensor locations, and the channel statistics, i.e., q_0 and q_1 , to localize the target. The following theorem characterizes the CRLB for this estimation problem.

Theorem VI.1: For the above formulated channel-aware estimation problem, the FIM $\mathbf{J}$ is given as follows [74]:

$$\mathbf{J} = \sum_{i=1}^N \sum_{\tilde{m}_i=0}^1 \frac{\nabla_{\theta} p(\tilde{m}_i|\theta) \nabla_{\theta}^T p(\tilde{m}_i|\theta)}{p(\tilde{m}_i|\theta)} \quad (32)$$

where

$$\nabla_{\theta} p(\tilde{m}_i|\theta) = \sum_{m_i} p(\tilde{m}_i|m_i) \nabla_{\theta} p(m_i|\theta) \quad (33)$$

$$\nabla_{\theta} p(m_i = l|\theta) \quad (34)$$

$$= \frac{\lambda_{il}}{2\sqrt{2\pi}\sigma} \left[\frac{d_i^{-\frac{n}{2}}}{\sqrt{P_0}}, n a_i d_i^{-2} (x_i - x_t), n a_i d_i^{-2} (y_i - y_t) \right]^T$$

and λ_{il} was defined in Theorem III.2.

B. Soft Decoding in Rayleigh Fading Channel With Coherent Reception

In this section, a soft-decoding link is considered. Let the channel between the i th sensor and the FC be a discrete-time Rayleigh flat fading channel with a stationary and ergodic complex gain, which is assumed to remain constant during a symbol transmission. Also, assume the channels to be independent of each other. With coherent reception at the receiver and unit power for fading, the observation model at the FC is given as

$$\tilde{m}_i = h_i m_i + v_i \quad (35)$$

where $v_i \sim \mathcal{N}(0, \sigma_v^2)$. Then, the conditional pdf $p(\tilde{m}_i|m_i)$ can be obtained as in [96]

$$p(\tilde{m}_i|m_i) = \frac{2\sigma_v}{\sqrt{2\pi}(1+2\sigma_v^2)} e^{\frac{-\tilde{m}_i^2}{2\sigma_v^2}} \times \left[1 + m_i \sqrt{2\pi} \alpha \tilde{m}_i e^{\frac{(\alpha \tilde{m}_i)^2}{2}} Q(-\alpha m_i \tilde{m}_i) \right] \quad (36)$$

where $\alpha = 1/(\sigma_v \sqrt{1+2\sigma_v^2})$.

In this case, the FC needs the phase of the received signal. The localization algorithm does not need to estimate the channel gain h_i , as the algorithm incorporates the channel statistics for estimation. The following theorem characterizes the CRLB for this estimation problem.

Theorem VI.2: For the channel-aware localization problem with soft decoding in Rayleigh fading channels with coherent reception, the FIM is provided as follows [74]:

$$\mathbf{J} = \sum_{i=1}^N \int_{-\infty}^{\infty} \frac{\nabla_{\theta} p(\tilde{m}_i | \theta) \nabla_{\theta}^T p(\tilde{m}_i | \theta)}{p(\tilde{m}_i | \theta)} d\tilde{m}_i \quad (37)$$

where $\nabla_{\theta} p(\tilde{m}_i | \theta)$ has the same form as (33), but with a different $p(\tilde{m}_i | m_i)$, provided by (36).

C. Soft Decoding in Rayleigh Fading Channel With Noncoherent Reception

Assuming the same wireless channel conditions and binary signaling as in Section VI-B, we now analyze the third link layer design scenario where the FC employs phase noncoherent reception. Assuming that the receiver employs the widely used and low-cost energy detection (ED), which is a noncoherent detection scheme [97], it is natural to utilize ON/OFF binary signaling scheme, and the sensors can remain silent for $m_i = 0$. Under these assumptions, the conditional pdf has been derived [74]

$$p(\tilde{m}_i | m_i = 0) = \frac{1}{2\sigma_v^2} e^{-\frac{\tilde{m}_i}{2\sigma_v^2}} \quad (38)$$

$$p(\tilde{m}_i | m_i = 1) = \frac{1}{1 + 2\sigma_v^2} e^{-\frac{\tilde{m}_i}{1 + 2\sigma_v^2}}.$$

Note that the FC does not need to estimate the received signal phase. For this problem, the FIM has almost the same form as in (37). The differences lie in $p(\tilde{m}_i | m_i)$ and that the range of integral for the case of noncoherent reception is from 0 to ∞ , instead of between $-\infty$ and ∞ [74].

In Fig. 7, the RMS errors of target location estimates for six different link layer designs are provided. For comparison

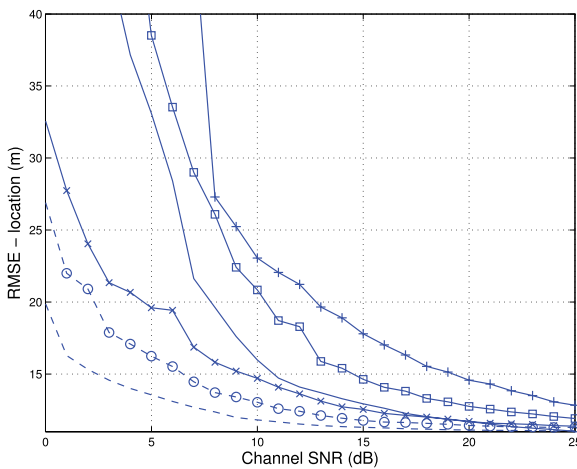


Fig. 7. RMS errors of different channel-aware estimation methods versus average channel SNRs. Dashed line: channel-aware coherent soft-decoding; dashed-circle line: channel-aware coherent hard-decoding; solid-cross line: channel-unaware coherent hard-decoding; solid line: channel-aware noncoherent soft-decoding; solid-square line: channel-aware noncoherent hard-decoding; solid-plus line: channel-unaware noncoherent hard-decoding [74].

purposes, for hard-decoding links, “channel-unaware” approaches are introduced that are unaware of the imperfection of the binary channels. From the figure, it is evident that the coherent receivers are superior in localization performance than the noncoherent receivers, irrespective of the decoding scheme and/or processing algorithm. Also observe that soft-decoding schemes always perform better than the corresponding hard-decoding schemes. Incorporating the channel statistics in the algorithm helps to use soft-decoding link designs and improve performance without increasing communication overhead.

VII. LOCALIZATION IN THE PRESENCE OF BYZANTINES

In this section, the adversarial setup is considered. One of the major security concerns in WSNs is the presence of Byzantines, which are malicious sensors that send corrupted data into the inference process [79]. In [81], Vempaty *et al.* studied the effect of such malicious sensors in the localization problem under the setup where all the sensors can communicate with the fusion center directly. The problem is addressed from both the attacker’s perspective by quantifying the optimal attacking strategy, and the network’s perspective by designing robust Byzantine mitigation schemes. Game-theory-based models have also been developed to analyze the interactions between the attacker and the network.

A. Optimal Attacking Strategy

A major fraction of the results in [81] has focused on the case when sensors send binary decisions and the Byzantines attack the network independently. In such an independent attack, each Byzantine sensor behaves independently and attacks the network based only on its own observation without any knowledge of the actions of other Byzantine sensors. The Byzantines’ strategy is characterized by p , the probability of flipping their quantized binary measurements before sending them to the FC. Let the observations received at the FC after undergoing potential corruption by Byzantines be represented as $\mathbf{u} = [u_1, \dots, u_N]^T$. Since the sensors are unaware of the estimator that is used at the FC, PCRLB and posterior/Bayesian FIM (as defined in Section II-B) are used as the metrics to analyze the estimation performance in the presence of Byzantines. Let α denote the fraction of the sensors in the network that are Byzantines, and the channels between the sensors and the FC are assumed to be ideal. For the adversarial Byzantines, the aim is to degrade the network performance at the FC to the maximum possible extent. This event of maximum damage when the FC becomes incapable of making any estimate from the local sensors’ data is referred to as blinding the FC. Mathematically speaking, this is when the data’s contribution to the posterior Fisher information matrix is zero. When this happens, the FC can only use its prior

information for location estimation. The condition for this to happen is given as follows.

Proposition VII.1: The data's contribution to posterior Fisher information $\mathbf{F}$ approaches 0 for $\alpha p = 1/2$ [81].

From this result, one can observe that to cause maximum degradation to the network with their constrained resources, Byzantines must use a high probability of flipping (p). Hence, the minimum fraction of Byzantines needed to blind the FC, $\alpha_{\text{blind}} = 1/2$ when $p = 1$. In other words, at least 50% of the sensors in the network must be Byzantines flipping their observations with $p = 1$ in order to blind the FC.

When α is lower than the blinding fraction, the Byzantines will try to make either $\text{tr}(\mathbf{F}^{-1})$ as large as possible or $|\mathbf{F}|$ as small as possible, thereby causing maximum damage. On the other hand, the FC would try to do the opposite, minimize $\text{tr}(\mathbf{F}^{-1})$ or maximize $|\mathbf{F}|$, thereby resulting in a zero-sum game. For this game, the sensor thresholds are their strategies η_B for Byzantines who are trying to maximize $\text{tr}(\mathbf{F}^{-1})$, and honest sensors adjust their threshold η_H to minimize $\text{tr}(\mathbf{F}^{-1})$ or maximize $|\mathbf{F}|$.

For example, consider a WSN with $N = 100$ sensors in a square region of 11×11 . The target is present at a location randomly distributed according to prior $p_0(\theta)$ with $\sigma_\theta = 2.1352$ such that its 99% confidence region covers the entire ROI. Let us assume that Byzantine sensors flip their observations with $p = 1$. Let the model parameters be $P_0 = 200$, $d_0 = 1$, and local sensor observations be corrupted by AWGN with standard deviation $\sigma = 3$. For $\alpha < \alpha_{\text{blind}} = 0.5$, as discussed before, there is a zero-sum game. The solution to this problem is given by saddle points. In Fig. 8, $|\mathbf{F}|$ is plotted with varying thresholds when $\alpha = 0.4$, and the saddle point in this setup is at $(8.5, 8.5)$. This result is expected as the best strategy for the Byzantines is to use the same strategy as the honest sensors and to always flip their decisions. Similar results have been obtained for different values of α [81].

Collaborative Attack: Previously, we discussed the case of independent Byzantine attacks. When these Byzantine

sensors collaboratively communicate with each other and attack in a coordinated fashion, it is referred to as the collaborative attack. A lower bound on α_{blind} for such an attack can be determined by considering the scenario where the colluding Byzantines know the true target location via collaboration, and use that information to attack the network [81].

B. Byzantine Mitigation Schemes

Several Byzantine mitigation schemes in WSNs that perform localization have been proposed, of which we discuss two specific ones. In the first scheme, Byzantines are identified as the sensors that have “abnormal” behavior [81]. In this work, such a behavior is quantified using the values of $\gamma \triangleq p(u_i = 1|\theta)$ for each sensor. The observed values are given by $\hat{\gamma}_i = p(u_i = 1|\hat{\theta})$, which are empirically estimated in an iterative manner, and $\hat{\gamma}_i$ at the $(T + 1)$ th iteration is

$$\hat{\gamma}_i(T+1) = \frac{T\hat{\gamma}_i(T) + u_i(T+1)}{T+1}. \quad (39)$$

If the expected values for honest and Byzantine sensors are $\hat{\gamma}_i^H = p(u_i = 1|\hat{\theta}^H, i = \text{Honest})$ and $\hat{\gamma}_i^B = p(u_i = 1|\hat{\theta}^B, i = \text{Byzantine})$, respectively, then the sensor is declared honest or Byzantine based on the test statistic

$$\Lambda_i(T) = \left| \frac{\hat{\gamma}_i(T) - \hat{\gamma}_i^H}{\hat{\gamma}_i(T) - \hat{\gamma}_i^B} \right|. \quad (40)$$

This test statistic is the ratio between the deviation of the i th sensor's estimated behavior from the honest sensor's expected behavior, and the deviation of the i th sensor's estimated behavior from the Byzantine sensor's expected behavior. This formulation is motivated by pattern recognition problems, where the type of the sensor is determined from the observed behavior.

Note that this scheme works for any fraction of Byzantines (α) in the network. While this scheme identifies almost all of the Byzantines, one major concern is regarding those sensors for which the test statistic $\Lambda_i(T)$ is close to 1. These sensors (termed as ambiguous) cannot be identified

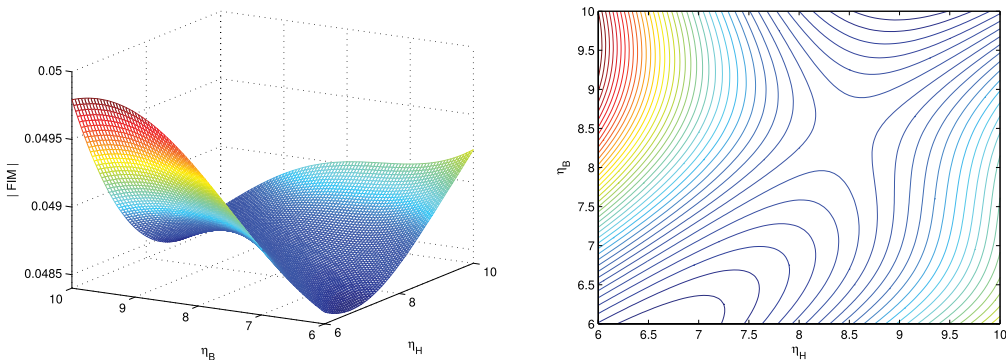


Fig. 8. Surface and contour plots of $|\mathbf{F}|$ versus honest and Byzantine sensor's threshold η_H and η_B [81].

reliably as Byzantine or honest, and a clear decision regarding the identity of the sensor cannot be made.

To handle the issue of ambiguous sensors that are at the boundary, Vempaty *et al.* [81] proposed a nonidentical dynamic quantizer design scheme. In such a scheme, the estimation model is sequential where at each iteration, all the sensors transmit their 1-bit decisions to the FC using the current thresholds. These thresholds are initialized to be the optimal identical thresholds. The FC makes an estimate of the target's location using this information and broadcasts this estimate to all the sensors. The sensors use this estimate to update their local thresholds in an adaptive manner. Therefore, the goal is now to design an appropriate functional form of the quantizers $\eta_H(\cdot)$ and $\eta_B(\cdot)$ for honest sensors and Byzantines, respectively.

These threshold functions can be designed by modeling this problem as a zero-sum game with $G[\eta^H(\cdot), \eta^B(\cdot)]$ as the objective function, where $G \triangleq |\mathbf{F}|$ denotes the determinant of the posterior FIM $\mathbf{F}$. While the FC tries to maximize the objective function, the Byzantines try to minimize it. When each sensor behaves independently, it has been shown that the posterior FI is maximized for $\eta^H(\hat{a}) = \hat{a}$. On the other hand, for Byzantines, it depends on whether they are in the blinding region or not. When $\alpha p < 1/2$, the Byzantines minimize this posterior FI by setting $\eta^B(\hat{a}) = \hat{a}$, which is expected since the Byzantines are flipping their observations. Note that when $\eta^B(\hat{a}) = \eta^H(\hat{a}) = \hat{a}$, the honest sensors send 0/1 with approximately equal probabilities.

When $\alpha p \geq 1/2$, mathematically the best strategy for the Byzantines is to use $\eta^B(\hat{a}) = \pm\infty$. In other words, they always send a 0 or a 1. This works in favor of the network, as any sensors always sending the same data can be easily identified by the FC using the scheme discussed before, and thereby their Byzantine actions can be overcome.

VIII. LOCALIZATION AS HIERARCHICAL CLASSIFICATION

In this section, we present a new paradigm of localization in WSNs that uses error-correcting codes by modeling localization as hierarchical classification [82]. In conventional approaches, the FC locates the target using an ML estimator or an MMSE estimator based on the received data. However, this approach has a high computational complexity because it performs optimization over the entire region of interest (ROI). Also, as shown in Section VII, these traditional approaches are prone to malicious attacks. While there are learning-based mitigation schemes for such scenarios as discussed in Section VII-B, another approach to mitigate Byzantines is to design schemes that are robust to such attacks. The scheme presented here using error-correcting codes provides multiple benefits of being computationally efficient and also more robust to Byzantine attacks. The basic intuition behind this approach is to formulate the localization problem as a hierarchical

classification problem. In this iterative approach, at each iteration, the FC divides the ROI into M subregions. Through M -ary hypothesis testing, the FC obtains a decision and declares the corresponding subregion as the ROI for the next iteration. In the M -ary hypothesis test, each sensor transmits binary quantized data to the FC using a predefined code matrix C . C is an $M \times N$ matrix, whose elements are $c_{(j+1),i} \in \{0,1\}$, $j = 0, \dots, M-1$ and $i = 1, \dots, N$. In the code matrix, its j th row corresponds to the j th subregion and its i th column corresponds to the i th sensor's binary decision rule. After collecting the binary decisions $\mathbf{u} = [u_1, u_2, \dots, u_N]^T$ from the local sensors, the FC makes an M -ary decision using the minimum Hamming distance criterion. In this manner, at each iteration, the target search space is reduced and the search stops if a prespecified stopping criterion is satisfied. At each iteration, the optimal division of the ROI depends on the network topology and the sensor distribution in the ROI, and it can be designed offline by k -means clustering [98], which provides Voronoi regions [99] with equal number of sensors in each subregion. For example, if the sensors are deployed uniformly in a ROI, the optimal division will be uniform as illustrated in Fig. 9.

Let us assume that N sensors are divided into M subregions. After each iteration, the number of sensors in the new ROI is reduced by a factor of M . After k iterations, there are N/M^k sensors in the ROI. Hence, at the $(k+1)$ th iteration, the code matrix C will have a size of $M \times N/M^k$.² Because the code matrix C should have more columns than rows, $k^{\text{stop}} < \log_M N$, in which k^{stop} denotes the number of iterations after which the algorithm stops. After k^{stop} iterations, only $N/M^{k^{\text{stop}}}$ sensors remain in the ROI, whose locations are averaged to get a coarse estimate of the target location, denoted as $\hat{\theta} = [\hat{x}_t, \hat{y}_t]^T$. In this iterative algorithm, the code matrix has to be designed at each iteration. At the k th iteration, let R_j^k denote the subregion corresponding to hypothesis H_j for $j = 0, 1, \dots, M-1$ and let S_j^k correspond to the set of sensors that are located inside R_j^k . Each sensor

²It is assumed that N is a multiple of M^k for $k = 0, 1, \dots, \log_M N - 1$.

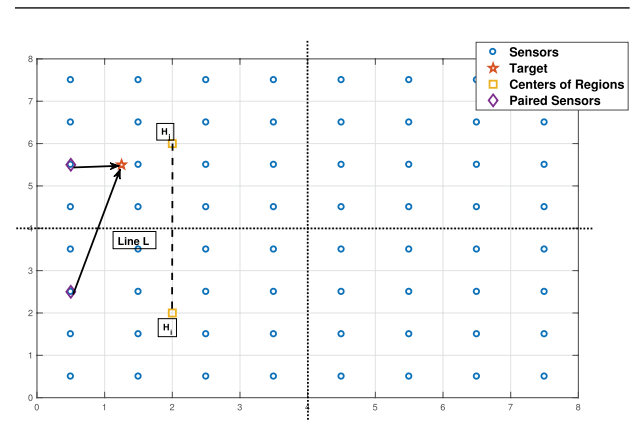


Fig. 9. ROI division example with a set of paired sensors.

$$C = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}.$$

i has a unique corresponding subregion, in which the sensor is located and the hypothesis associated with the subregion is denoted as $r^k(i)$. It is clear that $S_j^k = \{i \in \text{ROI}_k | r^k(i) = j\}$. When designing the code matrix, in the j th row, only those sensors (columns) that are in R_j^k should have 1's as their elements. Namely

$$c_{(j+1),i}^k = \begin{cases} 1, & \text{if } i \in S_j^k \\ 0, & \text{otherwise} \end{cases} \quad (41)$$

for $j = 0, \dots, M-1$ and $i \in \text{ROI}_k$. For example, for a WSN with $N = 16$ sensors symmetrically placed in a square grid with the $M = 4$ subregions as in Fig. 9, the code matrix is shown at the top of the page.

This can also be deemed as each sensor i adopting a threshold η_i^k for quantization. Let each subregion R_j^k correspond to a target location θ_j^k for $j = 0, 1, \dots, M-1$, which is set as the subregion's center. The performance of this scheme is evaluated using an analytically tractable metric, the target region's probability of detection, which is suitable when the goal is to identify the subregion in which the target is located. The main result about the asymptotic optimality of the approach relies on the following condition. Note that the approach is still applicable to WSNs when the condition is not satisfied.

Assumption VIII.1: For any target location $\theta \in R_j^k$ and any $0 \leq k \leq k^{\text{stop}}$, there is a bijection function f from S_j^k to S_l^k , where $0 \leq l \leq M-1$ and $l \neq j$, such that $f(i_j) = i_l$, $\eta_{i_j}^k = \eta_{i_l}^k$, and $d_{i_j} < d_{i_l}$, where $i_j \in S_j^k$, $i_l \in S_l^k$, and d_{i_j} (d_{i_l}) is the distance between sensor i_j (i_l) and θ .

In Fig. 9, an example WSN which satisfies this condition is provided. For each sensor $i_j \in S_j^k$, because of the symmetric ROI division, there is a symmetrically located sensor $i_l \in S_l^k$ as shown below: Join the two subregions' centers and draw a perpendicular bisector L as illustrated in Fig. 9. Sensor $i_l \in S_l^k$ and sensor i_j are symmetric with respect to line L . These sensors have the same thresholds. Namely, because of the sensors' symmetric deployment, $\eta_{i_j}^k = \eta_{i_l}^k$. Clearly, when $\theta \in R_j^k$, $d_{i_j} < d_{i_l}$. The asymptotic result is provided next.

Theorem VIII.2: Under Assumption VIII.1, $\lim_{N \rightarrow \infty} P_D = 1$ [82].

A. Enhancements to Address Practical Concerns

Enhancements to the basic coding scheme have been proposed to address practical concerns of WSNs. Two practical aspects considered are the presence of Byzantines discussed in the previous section, and the presence of nonideal communication channels.

1) *Exclusion Approach with Weighted Average:* To mitigate possible Byzantine attacks, at each iteration, the best two subregions are chosen for the next iteration. At the end of the iterations, the FC takes the weighted average of sensor locations to locate the target. In essence, the following modifications are made.

- After each iteration, the two best subregions are kept. After the k th iteration, the code matrix's size is $M \times (2^k N / M^k)$, and the total number of iterations has to satisfy $k^{\text{stop}} < \log_{M/2} N$.
- At the end of the iterations, the weighted average of the sensor locations is used as the target location estimate, with weights being the sensors' local binary decisions. A decision of $u_i = 1$ means that the target is near sensor i , and the weighted average makes sure that only the sensors close to the target contribute to the target's location estimate.

As mentioned before, an additional benefit of the approach is its robustness to attacks. The code-matrix-based localization approach ensures robustness to the presence of Byzantines due to code matrix's fault-tolerant capability. When the exclusion approach is used, a total of at most $(2^{k+1} N / M^{k+1}) - 1$ faults can be tolerated.

Proposition VIII.3: At the $(k+1)$ th iteration, the maximum fraction of Byzantines which can be dealt with is limited by $\alpha_f^k = 2/M - M^k / (2^k N)$ [82].

Theorem VIII.4: Let α denote the fraction of Byzantines in the network. With Assumption VIII.1, when $\alpha < 0.5$, $\lim_{N \rightarrow \infty} P_D = 1$ [82].

An example is illustrated in Fig. 9, in which $N = 512$ sensors are placed on a uniform 8×8 grid. At each sensor, the received signal is corrupted by AWGN with standard deviation $\sigma = 3$. The target's signal power at the reference distance is $P_0 = 200$. At each iteration, the ROI is divided into $M = 4$ equal subregions as illustrated in Fig. 9. After $k^{\text{stop}} = 2$ iterations, there are 32 sensors in the ROI. For a fair comparison, the exclusion approach stops after $k^{\text{stop}} = 4$ iterations with 32 sensors left in the ROI. In Fig. 10, three target location approaches, including the coding approach, the coding approach with exclusion, and the ML-based approach discussed in Section III, are compared in terms of their target localization performance in the presence of Byzantine attacks. According to Fig. 10, it is clear that when the fraction of Byzantine $\alpha \leq 0.375$, the coding approach with exclusion performs the best among the three. On the other hand, the ML-based approach outperforms the other two when $\alpha > 0.375$. This corroborates our theoretical

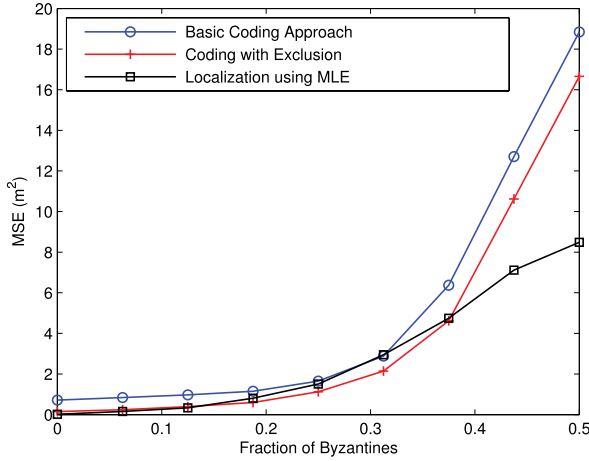


Fig. 10. Estimation performance of the three localization schemes: coding approach, coding with exclusion, and ML-based localization [82].

understanding that the coding-based approaches have a limit on the total number (fraction) of Byzantines (faults) that they can tolerate. The traditional approach has an upper limit of 0.5 for α [81], which the coding-based approaches achieve asymptotically. In practical scenarios, this limit is lower than 0.5.

2) *Soft-Decision Decoding for Fading Channels*: To combat the imperfect channel effects, at each iteration, a soft-decision decoding rule is used, instead of the minimum Hamming distance decoding rule discussed earlier. At each iteration, the sensors send their decisions $\mathbf{u}^k$, which may be modified by Byzantines, to the FC. Let us denote the received analog observations at the FC as $\mathbf{v}^k = [v_1^k, v_2^k, \dots, v_{N_k}^k]$, where the received data are related to the transmitted decisions as follows: $v_i^k = h_i^k(-1)^{u_i^k}\sqrt{E_b} + n_i^k$, $v_i \in \{1, \dots, N_k\}$, where h_i^k denotes the fading channel coefficient, E_b denotes the energy per channel bit, and n_i^k is the AWGN noise with variance σ_f^2 . Here, the channel coefficients are assumed to be Rayleigh distributed with variance σ_h^2 . The reliability for each of the received data is defined as follows:

$$\psi_i^k = \log \frac{p(v_i^k | u_i^k = 0)}{p(v_i^k | u_i^k = 1)} \quad (42)$$

for $i = 1, \dots, N_k$, in which $p(v_i^k | u_i^k)$ can be derived using the Rayleigh fading channel statistics. The F -distance is defined as $d_F(\boldsymbol{\psi}^k, \mathbf{c}_{j+1}^k) = \sum_{i=1}^{N_k} (\psi_i^k - (-1)^{c_{j+1,i}^k})^2$, where $\boldsymbol{\psi}^k = [\psi_1^k, \dots, \psi_{N_k}^k]$ and $\mathbf{c}_{j+1}^k$ denotes the j th row of the code matrix $\mathbf{C}^k$. With these definitions, the FC makes a decision by finding a subregion R_j^k such that $\mathbf{c}_j^k$ and $\boldsymbol{\psi}^k$ have the minimum F -distance. Under the assumption that for a fixed M , $(N/M^{k+1}) \rightarrow \infty$ as $N \rightarrow \infty$ for $k = 0, \dots, k^{\text{stop}}$, the asymptotic target detection performance of the soft-decision-decoding-based approach in the presence of Byzantine attacks is provided by the following theorem.

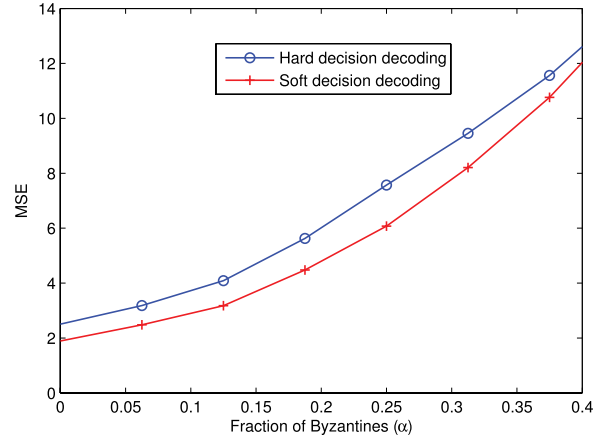


Fig. 11. Location estimation performance of the coding with exclusion approaches based on soft- and hard-decision decoding, respectively [82].

Theorem VIII.5: Under Assumption VIII.1, when $\alpha < 0.5$, $\lim_{N \rightarrow \infty} P_D = 1$ [82].

In the following example, the system parameters are set as follows: $N = 512$, $M = 4$, $A = 8^2 = 64$ sq. units, $P_0 = 200$, $\sigma = 3$, $E_b = 1$, $\sigma_f = 3$, and $E[(h_i^k)^2] = 1$, which corresponds to $\sigma_h^2 = 1 - (\pi/4)$. As shown in Fig. 11, the soft-decision-decoding-based approach outperforms the one based on hard-decision decoding. A detailed comparative study on accuracy performance and computational complexity was provided in [82].

IX. DISCUSSION AND CONCLUSION

In this paper, we have reviewed recent developments in RSS-based localization in wireless sensor networks. We have focused on several important practical issues and their solutions for such a localization problem. To reduce communication costs and sensor power consumption, local sensors send quantized measurements to the FC. It was shown that an ML estimator fusing quantized data can reach the CRLB even when both the number of quantization bits and the number of sensors are relatively small. Further, quantizer design schemes based on Fisher information and entropy help to make the estimation performance more robust, especially when a binary quantizer is used. To further conserve system resources, an iterative sensor selection approach was introduced to activate only the most informative sensors. At each iteration, target location information was measured using either PCRLB or mutual information. Sensor selection approaches based on these two measures provide similar estimation performance, but the PCRLB has a much smaller computational complexity, especially when the number of selected sensors is large.

For a resource constrained WSN with noisy and fading channels, channel-aware target localization was introduced, which incorporates channel statistics and decoding scheme

characteristics at the receiver. Three link layer designs were considered for channel-aware localization, including hard-decoding link modeled as a binary symmetric channel, Rayleigh fading coherent soft-decoding link, and Rayleigh fading non-coherent soft-decoding link. For both hard- and soft-decoding links, coherent reception approaches outperform noncoherent reception approaches. By adopting channel-aware target localization, it is possible to use soft-decoding links at the receiver to achieve better performance than corresponding hard-decoding links without extra communication costs.

For the scenario where malicious sensors exist in the system, the localization performance was characterized and mitigation schemes were presented. Two types of mitigation schemes were presented: learning based and coding-theory based. The coding-theory-based localization scheme was also shown to be computationally efficient and robust to channel errors.

Some interesting and challenging problems still need to be further investigated for RSS-based localization in WSNs. For example, in the problem of localizing multiple noncooperative targets where the signals from multiple targets are not well separable in a certain transformation domain (e.g., space, time, or frequency domain), the complexity of the ML estimator increases exponentially with the number of targets. The control of mobile agents, such as swarms of drones, over multiple time steps to achieve better localization/tracking performance involves an exponentially increasing complexity. Even though approaches dealing with harsh environments with multipath fading, obstruction, and anisotropic propagation media have been proposed, such as RSS fingerprinting, they rely on laborious offline building of radio maps and are sensitive to environment changes. Algorithms with lower complexity and that exhibit robustness for such harsh environments need to be developed. ■

REFERENCES

- [1] K. Pahlavan, X. Li, and J.-P. Mäkelä, "Indoor geolocation science and technology," *IEEE Commun. Mag.*, vol. 40, no. 2, pp. 112–118, Feb. 2002.
- [2] A. H. Sayed, A. Tarighat, and N. Khajehnouri, "Network-based wireless location: Challenges faced in developing techniques for accurate wireless location information," *IEEE Signal Process. Mag.*, vol. 22, no. 4, pp. 24–40, Jul. 2005.
- [3] H. Wymeersch, J. Lien, and M. Z. Win, "Cooperative localization in wireless networks," *Proc. IEEE*, vol. 97, no. 2, pp. 427–450, Feb. 2009.
- [4] N. Patwari and J. Wilson, "RF sensor networks for device-free localization: Measurements, models, and algorithms," *Proc. IEEE*, vol. 98, no. 11, pp. 1961–1973, Nov. 2010.
- [5] M. Z. Win, "Network localization and navigation via cooperation," *IEEE Commun. Mag.*, vol. 49, no. 5, pp. 56–62, May 2011.
- [6] D. Lymberopoulos and J. Liu, "The microsoft indoor localization competition: Experiences and lessons learned," *IEEE Signal Process. Mag.*, vol. 34, no. 5, pp. 125–140, Sep. 2017.
- [7] Y. Shen and M. Z. Win, "Fundamental limits of wideband localization—Part I: A general framework," *IEEE Trans. Inf. Theory*, vol. 56, no. 10, pp. 4956–4980, Oct. 2010.
- [8] Y. Shen, H. Wymeersch, and M. Z. Win, "Fundamental limits of wideband localization—Part II: Cooperative networks," *IEEE Trans. Inf. Theory*, vol. 56, no. 10, pp. 4981–5000, Oct. 2010.
- [9] D. Dardari, A. Conti, U. Finner, A. Giorgetti, and M. Z. Win, "Ranging with ultrawide bandwidth signals in multipath environments," *Proc. IEEE*, vol. 97, no. 2, pp. 404–426, Feb. 2009.
- [10] S. Bartoletti, W. Dai, A. Conti, and M. Z. Win, "A mathematical model for wideband ranging," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 2, pp. 216–228, Mar. 2015.
- [11] S. Mazuelas, A. Conti, J. Allen, and M. Win, "Soft range information for network localization," *IEEE Trans. Signal Process.*, to be published.
- [12] N. Patwari, J. N. Ash, S. Kyperountas, A. O. Hero, R. L. Moses, and N. S. Correal, "Locating the nodes: Cooperative localization in wireless sensor networks," *IEEE Signal Process. Mag.*, vol. 22, no. 4, pp. 54–69, Jul. 2005.
- [13] A. Conti, M. Guerra, D. Dardari, N. Decarli, and M. Z. Win, "Network experimentation for cooperative localization," *IEEE J. Sel. Areas Commun.*, vol. 30, no. 2, pp. 467–475, Feb. 2012.
- [14] M. F. Fallon and S. J. Godsill, "Acoustic source localization and tracking of a time-varying number of speakers," *IEEE Trans. Audio Speech Lang. Process.*, vol. 20, no. 4, pp. 1409–1415, May 2012.
- [15] O. Ozdemir, R. Niu, and P. K. Varshney, "Tracking in wireless sensor networks using particle filtering: Physical layer considerations," *IEEE Trans. Signal Process.*, vol. 57, no. 5, pp. 1987–1999, May 2009.
- [16] R. Niu, R. S. Blum, P. K. Varshney, and A. L. Droz, "Target localization and tracking in noncoherent multiple-input multiple-output radar systems," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 48, no. 2, pp. 1466–1489, Apr. 2012.
- [17] S. Mazuelas, Y. Shen, and M. Z. Win, "Belief condensation filtering," *IEEE Trans. Signal Process.*, vol. 61, no. 18, pp. 4403–4415, Sep. 2013.
- [18] J. Prieto, S. Mazuelas, and M. Z. Win, "Context-aided inertial navigation via belief condensation," *IEEE Trans. Signal Process.*, vol. 64, no. 14, pp. 3250–3261, Jun. 2016.
- [19] F. Meyer, P. Braca, P. Willett, and F. Hlawatsch, "A scalable algorithm for tracking an unknown number of targets using multiple sensors," *IEEE Trans. Signal Process.*, vol. 65, no. 13, pp. 3478–3493, Jul. 2017.
- [20] F. Meyer et al., "Message passing algorithms for scalable multitarget tracking," *Proc. IEEE*, vol. 106, no. 2, pp. 221–259, Feb. 2018.
- [21] Y. Shen, S. Mazuelas, and M. Z. Win, "Network navigation: Theory and interpretation," *IEEE J. Sel. Areas Commun.*, vol. 30, no. 9, pp. 1823–1834, Oct. 2012.
- [22] T. Wang, Y. Shen, A. Conti, and M. Z. Win, "Network navigation with scheduling: Error evolution," *IEEE Trans. Inf. Theory*, vol. 63, no. 11, pp. 7509–7534, Nov. 2017.
- [23] Z. Liu, W. Dai, and M. Z. Win, "Mercury: An infrastructure-free system for network localization and navigation," *IEEE Trans. Mobile Comput.*, vol. 17, no. 5, pp. 1119–1133, May 2018.
- [24] J. Liu, F. Zhao, and D. Petrovic, "Information-directed routing in ad hoc sensor networks," *IEEE J. Sel. Areas Commun.*, vol. 23, no. 4, pp. 851–861, Apr. 2005.
- [25] C. M. Kreucher, A. O. Hero, K. D. Kastella, and M. R. Morelande, "An information-based approach to sensor management in large dynamic networks," *Proc. IEEE*, vol. 95, no. 5, pp. 978–999, May 2007.
- [26] J. L. Williams, J. W. Fisher, III, and A. S. Willsky, "Approximate dynamic programming for communication-constrained sensor network management," *IEEE Trans. Signal Process.*, vol. 55, no. 8, pp. 4300–4311, Aug. 2007.
- [27] R. Tharmarasa, T. Kirubarajan, M. L. Hernandez, and A. Sinha, "PCRLB-based multisensor array management for multitarget tracking," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 43, no. 2, pp. 539–555, Apr. 2007.
- [28] L. Zuo, R. Niu, and P. K. Varshney, "Posterior CRLB based sensor selection for target tracking in sensor networks," in *Proc. Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, vol. 2, Honolulu, HI, USA, Apr. 2007, pp. 1041–1044.
- [29] L. Zuo, R. Niu, and P. Varshney, "A sensor selection approach for target tracking in sensor networks with quantized measurements," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process.*, Mar. 2008, pp. 2521–2524.
- [30] G. M. Hoffman and C. J. Tomlin, "Mobile sensor network control using mutual information methods and particle filters," *IEEE Trans. Autom. Control*, vol. 55, no. 1, pp. 32–47, Jan. 2010.
- [31] E. Masazade, R. Niu, and P. K. Varshney, "Dynamic bit allocation for object tracking

- in wireless sensor networks," *IEEE Trans. Signal Process.*, vol. 60, no. 10, pp. 5048–5063, Oct. 2012.
- [32] B. J. Julian, M. Angermann, M. Schwager, and D. Rus, "Distributed robotic sensor networks: An information-theoretic approach," *Int. J. Robot. Res.*, vol. 31, pp. 1134–1154, Aug. 2012.
- [33] Y. Shen, W. Dai, and M. Z. Win, "Power optimization for network localization," *IEEE/ACM Trans. Netw.*, vol. 22, no. 4, pp. 1337–1350, Aug. 2014.
- [34] S. Bartoletti, A. Giorgetti, M. Z. Win, and A. Conti, "Blind selection of representative observations for sensor radar networks," *IEEE Trans. Veh. Technol.*, vol. 64, no. 4, pp. 1388–1400, Apr. 2015.
- [35] W. Dai, Y. Shen, and M. Z. Win, "Energy-efficient network navigation algorithms," *IEEE J. Sel. Areas Commun.*, vol. 33, no. 7, pp. 1418–1430, Jul. 2015.
- [36] F. Meyer, H. Wymeersch, M. Frohle, and F. Hlawatsch, "Distributed estimation with information-seeking control in agent networks," *IEEE J. Sel. Areas Commun.*, vol. 33, no. 11, pp. 2439–2456, Nov. 2015.
- [37] S. Monica and G. Ferrari, "UWB-based localization in large indoor scenarios: Optimized placement of anchor nodes," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 2, pp. 987–999, Apr. 2015.
- [38] Z. Liu, W. Dai, and M. Z. Win, "Node placement for localization networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Paris, France, May 2017, pp. 1–6.
- [39] W. Dai, Y. Shen, and M. Z. Win, "A computational geometry framework for efficient network localization," *IEEE Trans. Inf. Theory*, vol. 64, no. 2, pp. 1317–1339, Feb. 2018.
- [40] X. Yang and R. Niu, "Adaptive sensor selection for nonlinear tracking via sparsity-promoting approaches," *IEEE Trans. Aerosp. Electron. Syst.*, to be published.
- [41] D. Li and Y. H. Hu, "Energy-based collaborative source localization using acoustic microsensor array," *EURASIP J. Appl. Signal Process.*, vol. 2003, no. 4, p. 985029, Apr. 2003.
- [42] T. Pham, B. M. Sadler, and H. Papadopoulos, "Energy-based source localization via ad-hoc acoustic sensor network," in *Proc. IEEE Workshop Stat. Signal Process. (SSP)*, Sep./Oct. 2003, pp. 387–390.
- [43] C. Meesookho, U. Mitra, and S. Narayanan, "On energy-based acoustic source localization for sensor networks," *IEEE Trans. Signal Process.*, vol. 56, no. 1, pp. 365–377, Jan. 2008.
- [44] M. Z. Rahman, G. C. Karmakar, and L. S. Dooley, "Passive source localization using power spectral analysis and decision fusion in wireless distributed sensor networks," in *Proc. Int. Conf. Inf. Technol. Coding Comput. (ITCC)*, Apr. 2005, pp. 260–264.
- [45] L. M. Kaplan, Q. Le, and N. Molnar, "Maximum likelihood methods for bearings-only target localization," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, May 2001, pp. 3001–3004.
- [46] X. Sheng and Y.-H. Hu, "Energy based acoustic source localization," in *Proc. 2nd Int. Symp. Inf. Process. Sensor Netw. (IPSN)*, Apr. 2003, pp. 285–300.
- [47] X. Shen and Y.-H. Hu, "Maximum likelihood multiple-source localization using acoustic energy measurements with wireless sensor networks," *IEEE Trans. Signal Process.*, vol. 53, no. 1, pp. 44–53, Jan. 2005.
- [48] Y. Oshman and P. Davidson, "Optimization of observer trajectories for bearings-only target localization," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 35, no. 3, pp. 892–902, Jul. 1999.
- [49] S. Haykin, *Array Signal Processing*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1984.
- [50] K. Yao, R. E. Hudson, C. W. Reed, D. Chen, and F. Lorenzelli, "Blind beamforming on a randomly distributed sensor array system," *IEEE J. Sel. Areas Commun.*, vol. 16, no. 8, pp. 1555–1567, Oct. 1998.
- [51] C. W. Reed, R. Hudson, and K. Yao, "Direct joint source localization and propagation speed estimation," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, Mar. 1999, pp. 1169–1172.
- [52] J. C. Chen, R. E. Hudson, and K. Yao, "A maximum-likelihood parametric approach to source localizations," in *Proc. IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, May 2001, pp. 3013–3016.
- [53] S. S. Ioushua, O. Yair, D. Cohen, and Y. C. Eldar, "Cascade: Compressed carrier and DoA estimation," *IEEE Trans. Signal Process.*, vol. 65, no. 10, pp. 2645–2658, May 2017.
- [54] C. L. Godara, "Application of antenna arrays to mobile communications, Part II: Beamforming and direction-of-arrival considerations," *Proc. IEEE*, vol. 85, no. 8, pp. 1195–1245, Aug. 1997.
- [55] F. Bellili, C. Elguet, S. B. Amor, S. Affes, and A. Stéphenne, "Code-aided DoA estimation from turbo-coded QAM transmissions: Analytical CRLBs and maximum likelihood estimator," *IEEE Trans. Wireless Commun.*, vol. 16, no. 5, pp. 2850–2865, May 2017.
- [56] Y. Wang and K. C. Ho, "TDOA positioning irrespective of source range," *IEEE Trans. Signal Process.*, vol. 65, no. 6, pp. 1447–1460, Mar. 2017.
- [57] N. Vankayalapati, S. Kay, and Q. Ding, "TDOA based direct positioning maximum likelihood estimator and the Cramer-Rao bound," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 50, no. 3, pp. 1616–1635, Jul. 2014.
- [58] D. Li, K. D. Wong, Y. H. Hu, and A. M. Sayeed, "Detection, classification, and tracking of targets," *IEEE Signal Process. Mag.*, vol. 19, no. 2, pp. 17–29, Mar. 2002.
- [59] S. Mazuelas, R. M. Lorenzo, A. Bahillo, P. Fernandez, J. Prieto, and E. J. Abril, "Topology assessment provided by weighted barycentric parameters in harsh environment wireless location systems," *IEEE Trans. Signal Process.*, vol. 58, no. 7, pp. 3842–3857, Jul. 2010.
- [60] J. Prieto, S. Mazuelas, A. Bahillo, P. Fernandez, R. M. Lorenzo, and E. J. Abril, "Adaptive data fusion for wireless localization in harsh environments," *IEEE Trans. Signal Process.*, vol. 60, no. 4, pp. 1585–1596, Apr. 2012.
- [61] P. Biswas, T.-C. Liang, K.-C. Toh, Y. Ye, and T.-C. Wang, "Semidefinite programming approaches for sensor network localization with noisy distance measurements," *IEEE Trans. Autom. Sci. Eng.*, vol. 3, no. 4, pp. 360–371, Oct. 2006.
- [62] R. M. Vaghefi, M. R. Gholami, R. M. Buehrer, and E. G. Ström, "Cooperative received signal strength-based sensor localization with unknown transmit powers," *IEEE Trans. Signal Process.*, vol. 61, no. 6, pp. 1389–1403, Mar. 2013.
- [63] S. Tomic, "RSS-based localization in wireless sensor networks using convex relaxation: Noncooperative and cooperative schemes," *IEEE Trans. Veh. Technol.*, vol. 64, no. 5, pp. 2037–2050, May 2015.
- [64] Y. Wen, X. Tian, X. Wang, and S. Lu, "Fundamental limits of RSS fingerprinting based indoor localization," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, Kowloon, Hong Kong, Apr. 2015, pp. 2479–2487.
- [65] Z. Wu, K. Fu, E. Jedari, S. R. Shuvra, R. Rashidzadeh, and M. Saif, "A fast and resource efficient method for indoor positioning using received signal strength," *IEEE Trans. Veh. Technol.*, vol. 65, no. 12, pp. 9747–9758, Dec. 2016.
- [66] X. Wang, L. Gao, and S. Mao, "CSI phase fingerprinting for indoor localization with a deep learning approach," *IEEE Internet Things J.*, vol. 3, no. 6, pp. 1113–1123, Dec. 2016.
- [67] R. Niu and P. K. Varshney, "Target location estimation in wireless sensor networks using binary data," in *Proc. 38th Annu. Conf. Inf. Sci. Syst. (CISS)*, Mar. 2004.
- [68] R. Niu and P. K. Varshney, "Target location estimation in sensor networks with quantized data," *IEEE Trans. Signal Process.*, vol. 54, no. 12, pp. 4519–4528, Dec. 2006.
- [69] N. Katenka, E. Levina, and G. Michailidis, "Robust target localization from binary decisions in wireless sensor networks," *Technometrics*, vol. 50, no. 4, pp. 448–461, Jan. 2008.
- [70] H. Shi, X. Lo, Y. Shang, and D. Ma, "Cramer-Rao bound analysis of quantized RSSI based localization in wireless sensor networks," in *Proc. Int. Conf. Parallel Distrib. Syst. (ICPADS)*, Jul. 2005, pp. 32–36.
- [71] D. D. Selvaratnam, I. Shames, J. H. Manton, and B. Ristic (Nov. 2017). "Source localisation using binary measurements." [Online]. Available: <https://arxiv.org/abs/1711.10153>
- [72] E. Masazade, R. Niu, P. K. Varshney, and M. Keskinöz, "Energy aware iterative source localization for wireless sensor networks," *IEEE Trans. Signal Process.*, vol. 58, no. 9, pp. 4824–4835, Sep. 2010.
- [73] C. Meesookho and S. Narayanan, "Distributed range difference based target localization in sensor network," in *Proc. Conf. Rec. 39th Asilomar Conf. Signals Syst. Comput.*, Oct. 2005, pp. 205–209.
- [74] O. Ozdemir, R. Niu, and P. K. Varshney, "Channel aware target localization with quantized data in wireless sensor networks," *IEEE Trans. Signal Process.*, vol. 57, no. 3, pp. 1190–1202, Mar. 2009.
- [75] H. K. D. Sarma and A. Kar, "Security threats in wireless sensor networks," in *Proc. 40th Annu. IEEE Int. Carnahan Conf. Secur. Technol.*, Oct. 2006, pp. 243–251.
- [76] B. Kailkhura, V. S. S. Nadendla, and P. K. Varshney, "Distributed inference in the presence of eavesdroppers: A survey," *IEEE Commun. Mag.*, vol. 53, no. 6, pp. 40–46, Jun. 2015.
- [77] M. Pajic, I. Lee, and G. J. Pappas, "Attack-resilient state estimation for noisy dynamical systems," *IEEE Trans. Control Netw. Syst.*, vol. 4, no. 1, pp. 82–92, Mar. 2017.
- [78] X. Guo, A. S. Leong, and S. Dey, "Estimation in wireless sensor networks with security

- constraints," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 2, pp. 544–561, Apr. 2017.
- [79] A. Vempaty, L. Tong, and P. Varshney, "Distributed inference with Byzantine data: State-of-the-art review on data falsification attacks," *IEEE Signal Process. Mag.*, vol. 30, no. 5, pp. 65–75, Sep. 2013.
- [80] L. Zhang, G. Ding, Q. Wu, Y. Zou, Z. Han, and J. Wang, "Byzantine attack and defense in cognitive radio networks: A survey," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 3, pp. 1342–1363, Apr. 2015.
- [81] A. Vempaty, O. Ozdemir, K. Agrawal, H. Chen, and P. K. Varshney, "Localization in wireless sensor networks: Byzantines and mitigation techniques," *IEEE Trans. Signal Process.*, vol. 61, no. 6, pp. 1495–1508, Jun. 2013.
- [82] A. Vempaty, Y. S. Han, and P. K. Varshney, "Target localization in wireless sensor networks using error correcting codes," *IEEE Trans. Inf. Theory*, vol. 60, no. 1, pp. 697–712, Jan. 2014.
- [83] X. Li, "RSS-based location estimation with unknown pathloss model," *IEEE Trans. Wireless Commun.*, vol. 5, no. 12, pp. 3626–3633, Dec. 2006.
- [84] S. Mazuelas et al., "Robust indoor positioning provided by real-time RSSI values in unmodified WLAN networks," *IEEE J. Sel. Topics Signal Process.*, vol. 3, no. 5, pp. 821–831, Oct. 2009.
- [85] L. Kinsler and A. Frey, *Fundamentals of Acoustics*. New York, NY, USA: Wiley, 1962.
- [86] N. Levanon, *Radar Principles*. New York, NY, USA: Wiley, 1988.
- [87] H. L. van Trees, K. L. Bell, and Z. Tian, *Detection Estimation and Modulation Theory, Part I: Detection, Estimation, and Filtering Theory*, 2 ed. Hoboken, NJ, USA: Wiley, 2013.
- [88] H. L. van Trees and K. Bell, *Bayesian Bounds for Parameter Estimation and Nonlinear Filtering/Tracking*. New York, NY, USA: Wiley, 2007.
- [89] D. S. Lee and N. K. K. Chia, "A particle algorithm for sequential Bayesian parameter estimation and model selection," *IEEE Trans. Signal Process.*, vol. 50, no. 2, pp. 326–336, Feb. 2002.
- [90] K. J. Hintz and E. S. Mcvey, "Multi-process constrained estimation," *IEEE Trans. Syst. Man Cybern.*, vol. 21, no. 1, pp. 237–244, Jan. 1991.
- [91] F. Zhao, J. Shin, and J. Reich, "Information-driven dynamic sensor collaboration," *IEEE Signal Process. Mag.*, vol. 19, no. 2, pp. 61–72, Mar. 2002.
- [92] F. Zhao, J. Liu, J. Liu, L. Guibas, and J. Reich, "Collaborative signal and information processing: An information-directed approach," *Proc. IEEE*, vol. 91, no. 8, pp. 1199–1209, Aug. 2003.
- [93] C. Kreucher, K. Kastella, and A. O. Hero, III, "Sensor management using an active sensing approach," *Signal Process.*, vol. 85, no. 3, pp. 607–624, Mar. 2005.
- [94] L. Zuo, R. Niu, and P. Varshney, "Conditional posterior Cramér-Rao lower bounds for nonlinear sequential Bayesian estimation," *IEEE Trans. Signal Process.*, vol. 59, no. 1, pp. 1–14, Jan. 2011.
- [95] B. Chen, L. Tong, and P. K. Varshney, "Channel-aware distributed detection in wireless sensor networks," *IEEE Signal Process. Mag.*, vol. 23, no. 4, pp. 16–26, Jul. 2006.
- [96] R. Niu, B. Chen, and P. K. Varshney, "Fusion of decisions transmitted over Rayleigh fading channels in wireless sensor networks," *IEEE Trans. Signal Process.*, vol. 54, no. 3, pp. 1018–1027, Mar. 2006.
- [97] D. Tse and P. Viswanath, *Fundamentals of Wireless Communication*. New York, NY, USA: Cambridge Univ. Press, 2005.
- [98] P. Berkhin, "Survey of clustering data mining techniques," *Accrue Software*, San Jose, CA, USA, Tech. Rep., 2002.
- [99] F. Aurenhammer, "Voronoi diagram—A survey of a fundamental geometric data structure," *ACM Comput. Surv.*, vol. 23, no. 3, pp. 345–405, Sep. 1991.

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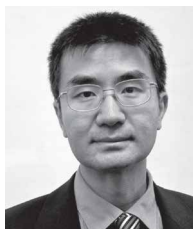
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